

Mathematics Bootcamp

Properties of Preferences and Utility Functions

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Today

- Indifference as an Equivalent Relation.
- Utility Function Representation
- Giving Some Structure to Preferences
 - Monotonicity
 - Local Non-Satiation
 - Convexity
 - Continuity

Preference Relations

- A **preference relation** $\succeq$ is a binary relation on X , that is, a subset of $X \times X$ that describes how the DM ranks any two elements in X .
- $x \succeq y$ reads: x is *at least as good* as y (x is *weakly better* than y).
- From $\succeq$ we derive:
 - Strict preference: $x \succ y \iff (x \succeq y \text{ and not } y \succeq x)$.
 - Indifference: $x \sim y \iff (x \succeq y \text{ and } y \succeq x)$.

Properties of Indifference

Indifference Relation

If $\succeq$ is rational (i.e., complete and transitive), then the induced indifference relation $\sim$ is:

1. **Reflexive:** $x \sim x$ for all $x \in X$.
2. **Symmetric:** $x \sim y \implies y \sim x$.
3. **Transitive:** $x \sim y \wedge y \sim z \implies x \sim z$.

Equivalence Relation

An equivalence relation is a binary relation that is reflexive, symmetric and transitive.

Equivalence Classes

Equivalence Classes

An equivalence relation induces a **partition** of the set: mutually disjoint subsets whose union equals the entire set.

Note that the indifference relation $\sim$

1. Is an equivalence relation.
2. It generates equivalence classes of indifferent elements.

For each $x \in X$, the equivalence class of x is denoted by

$$[x] = \{y \in X \mid x \sim y\}.$$

Equivalence Classes and Utility

Utility representation

A function $u : X \rightarrow \mathbb{R}$ *represents* $\succeq$ if

$$x \succeq y \iff u(x) \geq u(y).$$

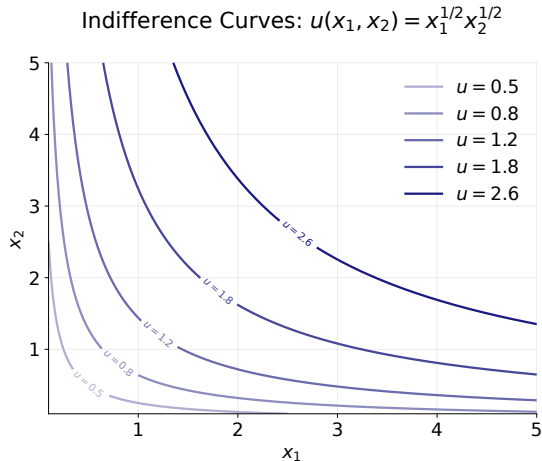
Corollary.

A function $u : X \rightarrow \mathbb{R}$ *represents* $\sim$ if

$$x \sim y \iff u(x) = u(y).$$

- Thus, all the elements in a equivalence class are mapped to the **same utility**!

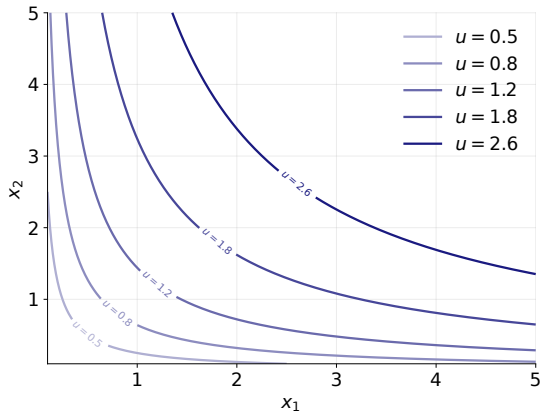
Indifference Curves



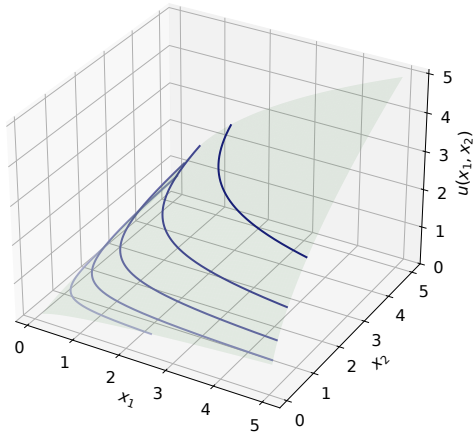
- Indifference curves are equivalence classes of X that map to the same utility in $\mathbb{R}$.
- Equivalence classes are disjoint, so indifference curves can never cross.

Utility Representation

Indifference Curves: $u(x_1, x_2) = x_1^{1/2} x_2^{1/2}$



Utility Surface + Indifference Curves: $u(x_1, x_2) = x_1^{1/2} x_2^{1/2}$



Useful Properties of Preferences

- In addition to rationality, we need more structure in preference relations. That is, we need to assume more properties.
- These properties have implications for:
 - Their utility function representation.
 - Their indifference curves.
- The best way to understand and learn them is by practicing proofs, many times.
 - You will have plenty of Weekly Problem Sets where you will do exactly that.

Represent Commodities with Ordered Lists

2.B Commodities

The decision problem faced by the consumer in a market economy is to choose consumption levels of the various goods and services that are available for purchase in the market. We call these goods and services *commodities*. For simplicity, we assume that the number of commodities is finite and equal to L (indexed by $\ell = 1, \dots, L$).

As a general matter, a *commodity vector* (or *commodity bundle*) is a list of amounts of the different commodities,

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_L \end{bmatrix},$$

and can be viewed as a point in $\mathbb{R}^L$, the *commodity space*.¹

We can use commodity vectors to represent an individual's consumption levels. The ℓ th entry of the commodity vector stands for the amount of commodity ℓ consumed. We then refer to the vector as a *consumption vector* or *consumption bundle*.

- **Notation:**

$x \geq y$ and $x \neq y$ imply that $x_l > y_l$ for some $l \in \{1, \dots, L\}$.

Monotonicity: More is Better

Definitions

A preference relation $\succeq$ is

weakly monotone if $x \geq y$ implies $x \succeq y$.

strictly monotone if $x \geq y$ and $x \neq y$ imply $x \succ y$.

A function $f : \mathbb{R}^L \rightarrow \mathbb{R}$ is

non-decreasing if $x \geq y$ implies $f(x) \geq f(y)$ for all $x, y \in \mathbb{R}^L$;

strictly increasing if $x \geq y$ and $x \neq y$ implies $f(x) > f(y) \forall x, y \in \mathbb{R}^L$.

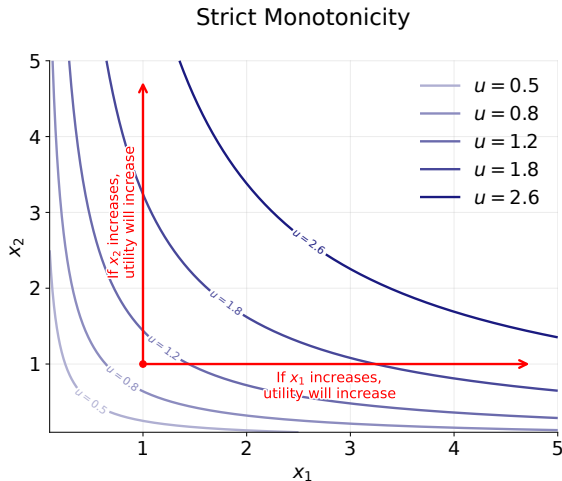
Proposition

If u represents $\succeq$ then

$\succeq$ is weakly monotone *iff* u is non-decreasing;

$\succeq$ is strictly monotone *iff* u is strictly increasing.

Monotonicity: More is Better



Local Non-Satiation: No Thick Indifference Curves

Definition: Local Nonsatiation

A preference relation $\succeq$ on X is *locally nonsatiated* if for every $x \in X$ and every $\varepsilon > 0$, there exists $y \in X$ such that $\|x - y\| \leq \varepsilon$ and $y \succ x$.

Thus, preferences are locally nonsatiated if for each x you can find an arbitrarily close y that is strictly better.

Recall that we are in a Euclidean space, so

$$\|x - y\| = \sqrt{\sum_{i=1}^L (x_i - y_i)^2}.$$

Local nonsatiation rules out “thick” indifference sets.

Local Non-Satiation: No Thick Indifference Curves

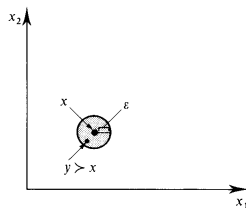
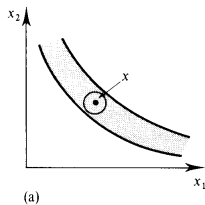
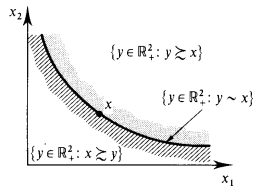


Figure 3.B.1
The test for local
nonsatiation.



(a)



(b)

Figure 3.B.2
(a) A thick indifference
set violates local
nonsatiation.
(b) Preferences
compatible with local
nonsatiation.

Monotonicity and Local Non-Satiation

Proposition

If $\succeq$ is strictly monotone, then $\succeq$ is locally non-satiated.

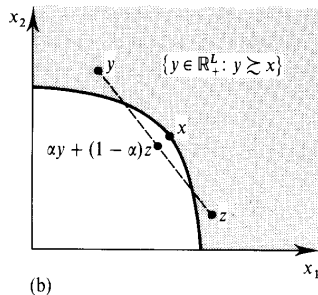
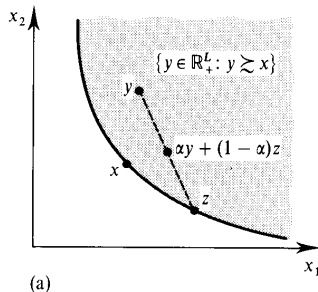
Proof. Suppose $\succeq$ is strictly monotone, i.e. $x \geq y$ and $x \neq y$ imply $x \succ y$. For each $x \in X$ and every $\varepsilon > 0$, let $y = x + \frac{\varepsilon}{L}\mathbf{1}$. Then

$$\|x - y\| = \sqrt{\sum_{i=1}^L \left(\frac{\varepsilon}{L}\right)^2} = \sqrt{L \left(\frac{\varepsilon}{L}\right)^2} = \frac{\varepsilon}{\sqrt{L}} \leq \varepsilon$$

Strict monotonicity implies that $y \succ x$.

However, local non-satiation does not imply monotonicity.

Convexity: Taste for Variety



Definition 3.B.4: The preference relation $\succsim$ on X is *convex* if for every $x \in X$, the upper contour set $\{y \in X : y \succsim x\}$ is convex; that is, if $y \succsim x$ and $z \succsim x$, then $\alpha y + (1 - \alpha)z \succsim x$ for any $\alpha \in [0, 1]$.

Figure 3.B.3(a) depicts a convex upper contour set; Figure 3.B.3(b) shows an upper contour set that is not convex.

Convexity: Taste for Variety

Definitions

A set $X \subseteq \mathbb{R}^L$ is

convex if for all $x, y \in X$, then $\alpha x + (1 - \alpha)y \in X \quad \forall \alpha \in [0, 1]$.

A preference relation $\succeq$ is

convex if $y \succeq x, z \succeq x$ imply $\alpha y + (1 - \alpha)z \succeq x \quad \forall \alpha \in [0, 1]$;

hence, for $x \in X$, the upper contour set, $\{y \in X : y \succeq x\}$ is convex.

str. convex if $y \succeq x, z \succeq x, y \neq z$ imply $\alpha y + (1 - \alpha)z \succ x \quad \forall \alpha \in (0, 1)$.

Implication: Indifference Curves

If $\succeq$ is *convex*, there is a preference for mixtures over extremes:

- Indifference curves being “convex to the origin” means they are bowed inward (not outward).

Level Sets and Quasiconcavity

Definitions

A function $f : X \rightarrow \mathbb{R}$ is

quasiconcave if $f(\alpha x + (1 - \alpha)y) \geq \min\{f(x), f(y)\} \forall \alpha \in [0, 1]$.

strictly quasiconcave if, for $x \neq y$ and $\alpha \in (0, 1)$,

$$f(\alpha x + (1 - \alpha)y) > \min\{f(x), f(y)\}.$$

A More Transparent Definition

A function $f : X \rightarrow \mathbb{R}$ is *quasiconcave* if every upper level set of f is convex. That is, $\{x \in X \mid f(x) \geq c\}$ is convex for every $c \in \mathbb{R}$.

Convexity: Taste for Variety

Convex Preferences imply Quasiconcave Utility

- If preferences $\succeq$ on X are convex, then mixtures are preferred to extremes.
- Then, every upper contour set $\{y \in X : y \succeq x\}$ is convex.
- Consequently, any utility representation $u : X \rightarrow \mathbb{R}$ is quasiconcave.

A Glimpse of Topology

Definitions

- A *neighborhood* of a point $x \in \mathbb{R}$ is any set V that contains an ε -neighborhood $V_\varepsilon(x) := (x - \varepsilon, x + \varepsilon)$ for some $\varepsilon > 0$.
- A set $G \subset \mathbb{R}$ is *open* if for each $x \in G$ there exists an ε such that $V_\varepsilon(x) \subset G$.
- A set $F \subset \mathbb{R}$ is *closed* if its complement $C(F) := \mathbb{R} \setminus F$ is open.

Implication

- Open sets do not contain their boundary points.
- Closed sets, by definition, include their boundary points.

Lower and Upper Contour Sets

- Now consider the *Upper Contour Set* and the *Lower Contour Set* of a preference relation:

$$U(x) := \{y \in X \mid y \succeq x\} \quad L(x) := \{y \in X \mid x \succeq y\}.$$

- Notice how the *indifference points* are both the *boundary points* and the *intersection* of both contour sets.
- If contour sets are closed, boundary points are included. That is, there are no gaps in the indifference curves.
- This is one of the ways in which we can characterize continuous preferences (see Kreps 1.14c in the next slide).

Continuity: No Gaps

Proposition 1.14

Continuity of preferences $\succeq$ on $\mathbb{R}_+^k$ imply the following, and any one of the following imply that preferences $\succeq$ on $\mathbb{R}_+^k$ are continuous. (Therefore, continuity of preferences could equivalently be defined by any one of the following, each of which implies all the others.)

- a. If $\{x_n\}$ is a sequence from $\mathbb{R}_+^k$ with $x_n \succeq y$ for all n , and if $\lim_{n \rightarrow \infty} x_n = x$, then $x \succeq y$. If $\{x_n\}$ is a sequence from $\mathbb{R}_+^k$ with $y \succeq x_n$ for all n , and if $\lim_{n \rightarrow \infty} x_n = x$, then $y \succeq x$.*
- b. If $\{x_n\}$ is a sequence from $\mathbb{R}_+^k$ with $\lim_{n \rightarrow \infty} x_n = x$, and if $x \succ y$, then for all sufficiently large n , $x_n \succ y$. And if $\lim_{n \rightarrow \infty} x_n = x$, and $y \succ x$, then for all sufficiently large n , $y \succ x_n$.*

Continuity: No Gaps

Proposition 1.14 (continued)

- c. For all $x \in \mathbb{R}_+^k$, the sets $\text{NBT}(x)$ and $\text{NWT}(x) = \{y \in \mathbb{R}_+^k : y \succeq x\}$ are both closed sets. (NWT is a mnemonic for No Worse Than.)*
- d. For all $x \in \mathbb{R}_+^k$, the sets $\text{SBT}(x) = \{y \in \mathbb{R}_+^k : y \succ x\}$ and $\text{SWT}(x) = \{y \in \mathbb{R}_+^k : x \succ y\}$ are both (relatively, in $\mathbb{R}_+^k$) open sets.⁴ (SBT is a mnemonic for Strictly Better Than, and SWT stands for Strictly Worse Than.)*

Debreu's Representation Theorem

Proposition 1.15

If $X = \mathbb{R}_+^k$ and preferences $\succeq$ are complete, transitive, and continuous on X , then $\succeq$ can be represented by a utility function u ; that is, $u(x) \geq u(y)$ if and only if $x \succeq y$.

- Previously we had seen a Necessary Condition for preferences to be represented by a utility function: Rationality.
 - Note that discontinuous preferences and rational can have a utility representation (continuous or not).
- With Rationality and **Continuity**, we have a Sufficient Condition for the existence of a continuous utility function representation.

Proposition 2.9 (Debreu's Theorem)

If a continuous function u represents $\succeq$, then $\succeq$ is continuous. Conversely, if $\succeq$ is continuous, it has a continuous utility representation.⁵